

ITAI 13: Intro to Machine Learning

Midterm

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Midterm Project Proposal

1. Dataset Overview and Source

- **Dataset Name:** Connecticut Real Estate Sales (2001–2021)
- **Dataset Source:** Kaggle
- **Original Dataset URL:** <https://www.kaggle.com/datasets/nrng19/real-estate>

Dimensions: Approximately 25,000 total records and 10 initial features.

2. Project Scope and Objectives

The objective of this project is to execute a rigorous data cleaning and exploratory data analysis (EDA) pipeline using Python within Jupyter Lab. To ensure a valid machine learning workflow and prevent data leakage, the raw dataset is programmatically split into a 70% training set (17,500 records) and a 30% testing set (7,500 records) prior to any manipulation[cite: 1, 3]. All preprocessing tasks—including handling missing values, managing invalid numerical entries, log normalization, standard scaling, and One-Hot Encoding—are performed exclusively on the training data while the test set remains entirely untouched[cite: 1, 3].

3. Machine Learning Proposal for Finals

- **Problem Type:** Supervised Learning (Regression)
- **Target Variable:** Sale Amount
- **Project Goal:** For the upcoming final project, our team will utilize the preprocessed training dataset to build, train, and evaluate multiple machine learning regression models (such as Linear Regression, Decision Trees, or Random Forests)[cite: 1, 3]. We aim to accurately predict real estate property sale prices based on key property and transaction characteristics, including assessed values, property types, residential types, geographic locations (towns), and recorded sale timelines.

